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Lee et al.

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(54) **DEVICE FOR COOLING A MOTOR**

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(30) **Foreign Application Priority Data**

Apr. 11, 2022 (KR) 10-2022-0044292

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H02K 9/19 (2006.01)

(52) **U.S. Cl.**
CPC ***H02K 1/20*** (2013.01); ***H02K 9/19*** (2013.01)

(58) **Field of Classification Search**
CPC .. H02K 1/20; H02K 9/00; H02K 9/12; H02K 9/19
USPC 310/52, 54, 58, 59
See application file for complete search history.

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(57) **ABSTRACT**

A device for cooling a motor can improve motor cooling performance with multiple cooling channels formed in a stator core of the motor, allowing a coolant to flow in both directions therethrough. The device is capable of maximizing motor cooling performance in such a manner that two or more multiple cooling channels are formed in a stator core with a separate coolant inlet and outlet. The coolant moves in a straight line in different directions along each cooling channel to cool the stator core, and thus the temperature gradient of the coolant is minimized to cool the entire area of the motor evenly.

14 Claims, 11 Drawing Sheets

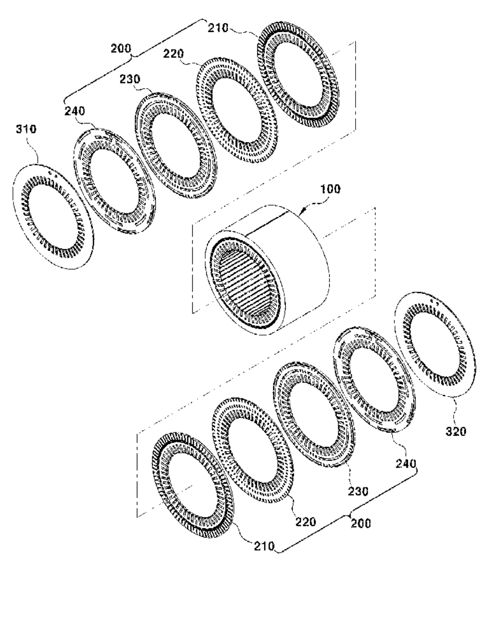

FIG. 1 – Prior Art

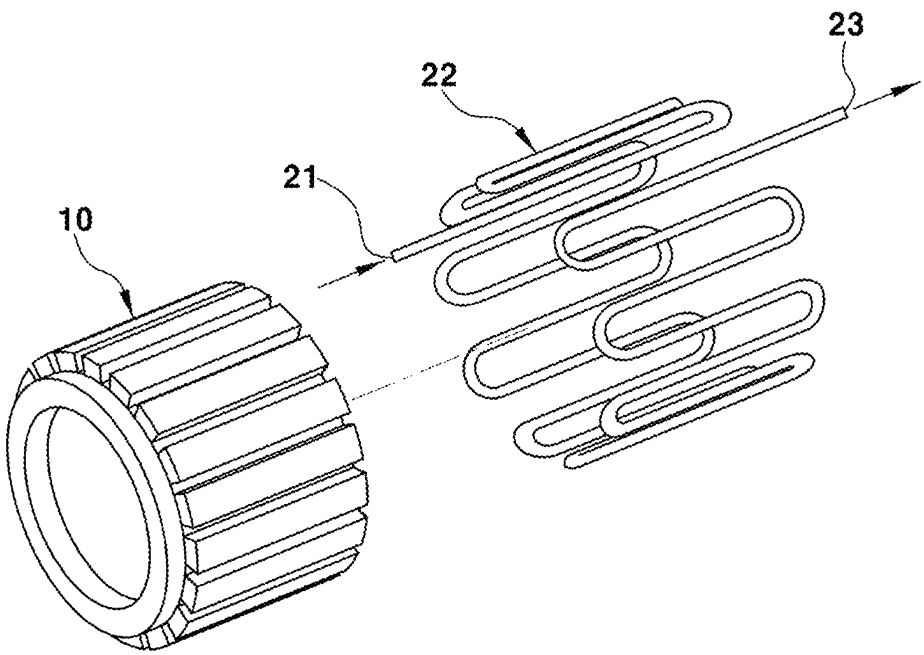

FIG. 2

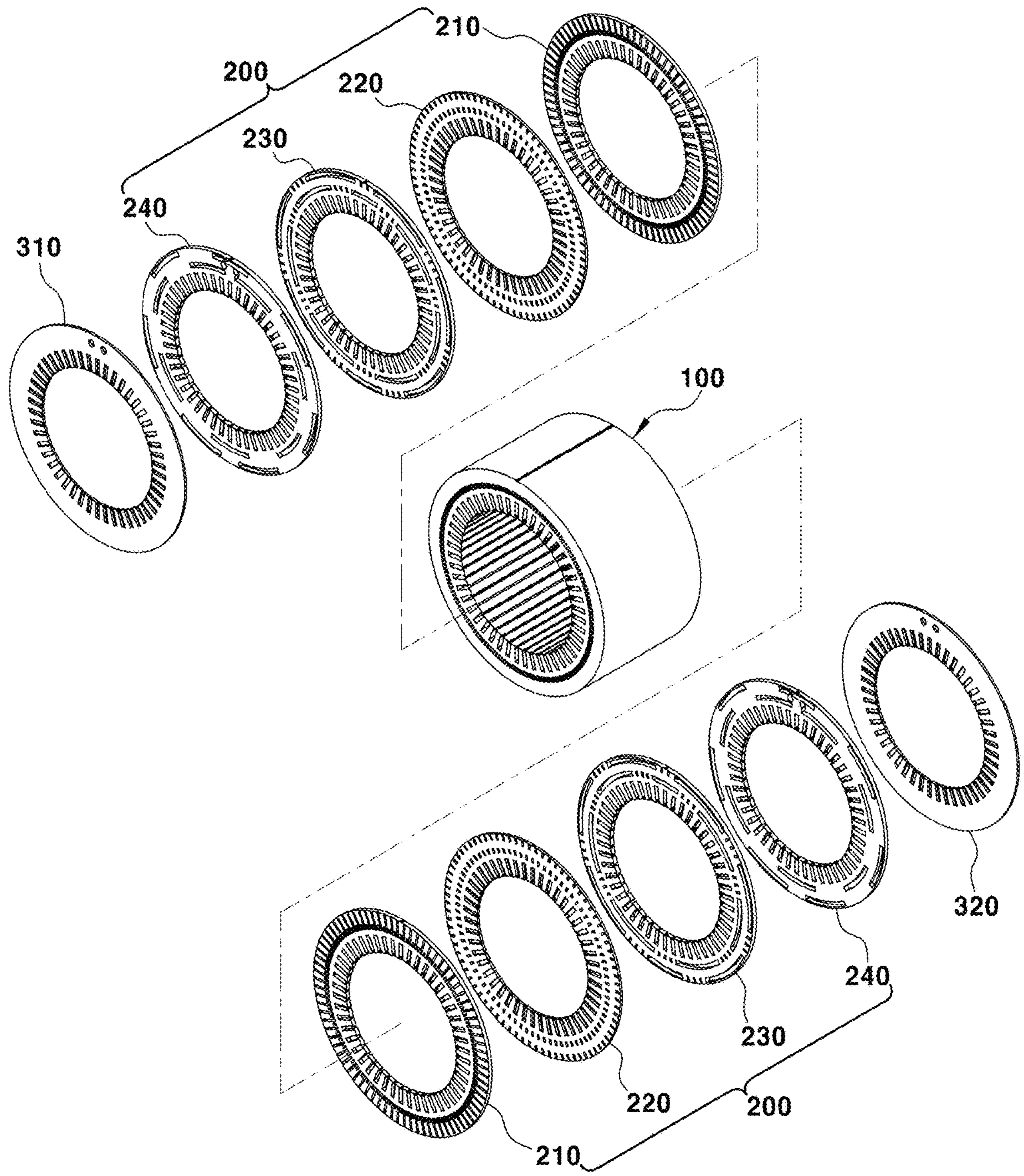

FIG. 3

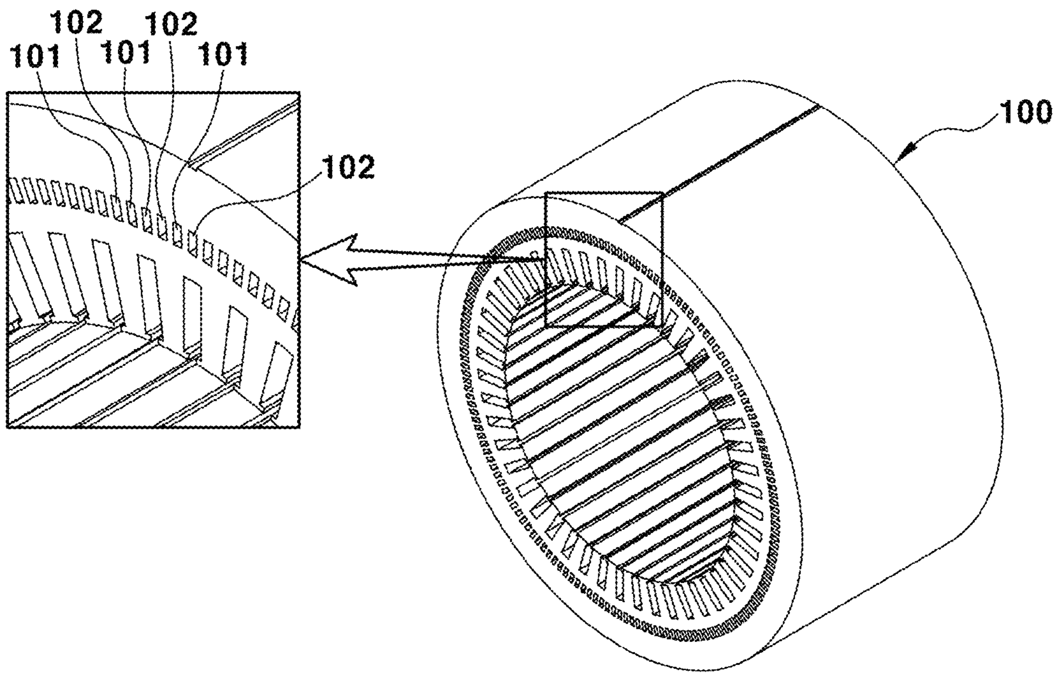

FIG. 4A

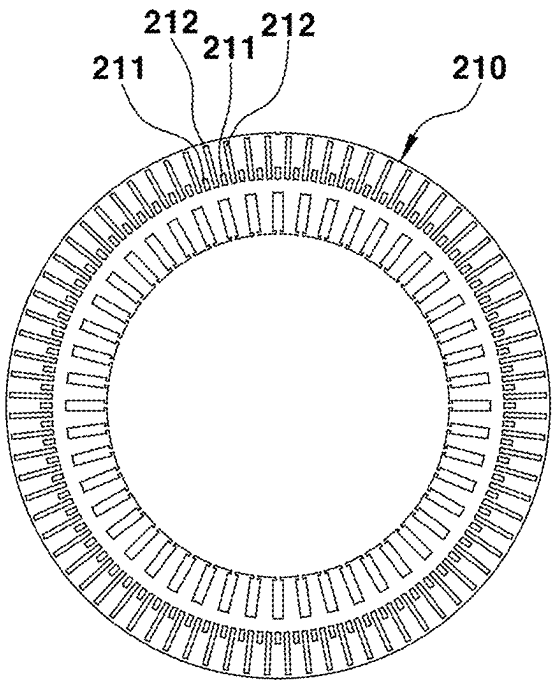

FIG. 4B

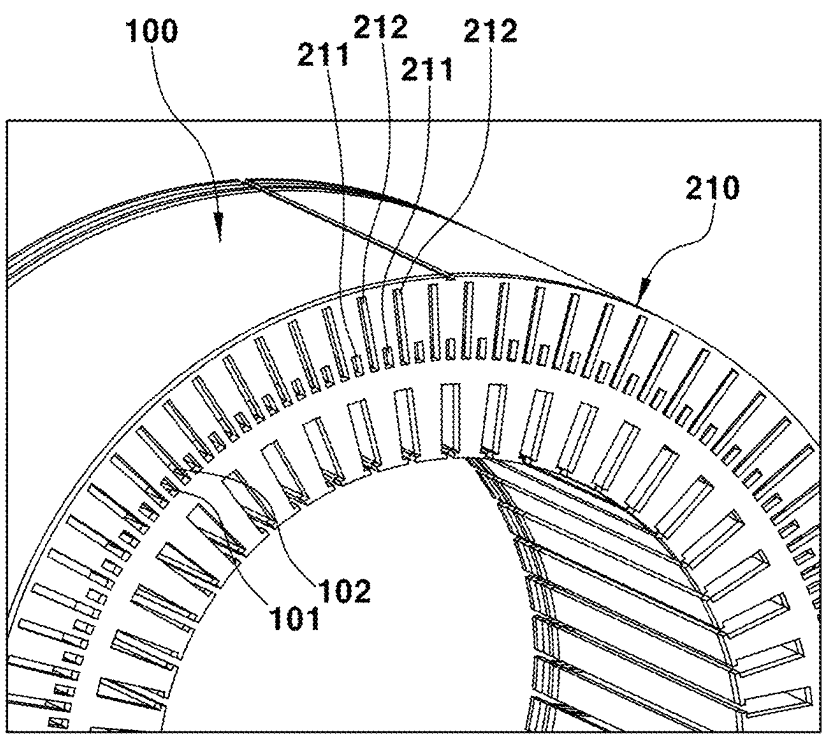

FIG. 5A

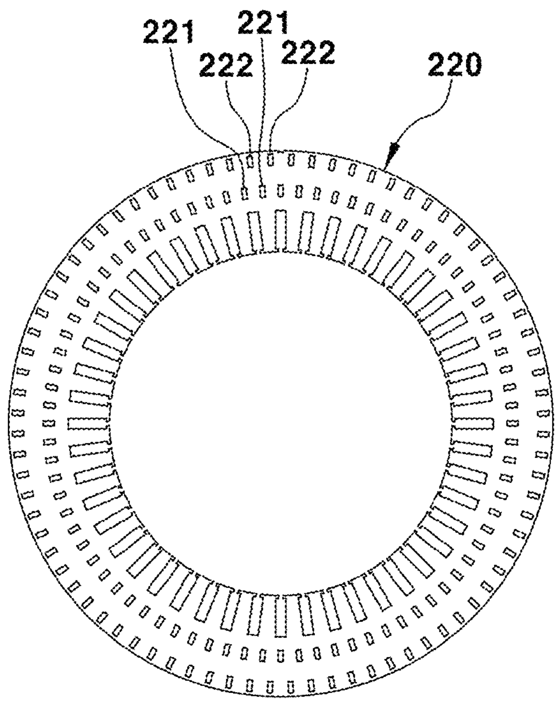

FIG. 5B

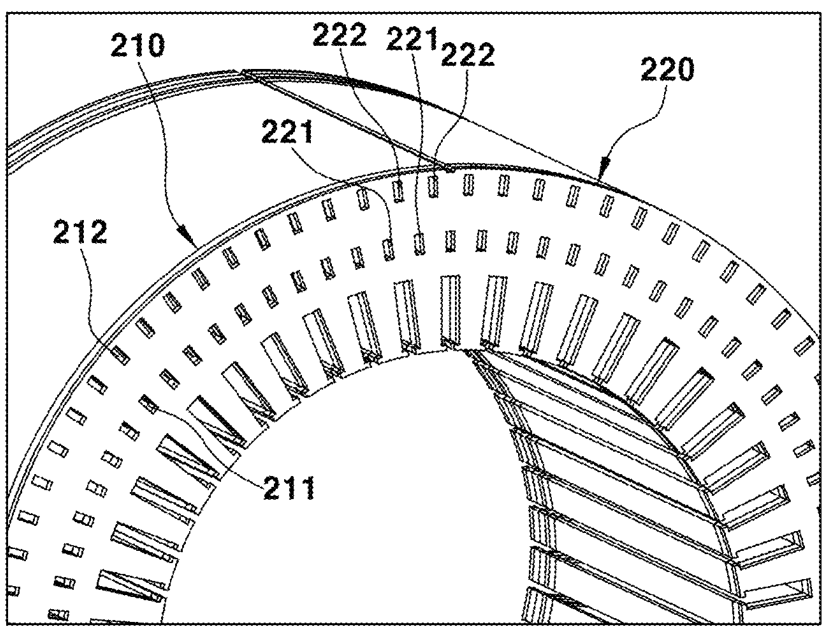

FIG. 6A

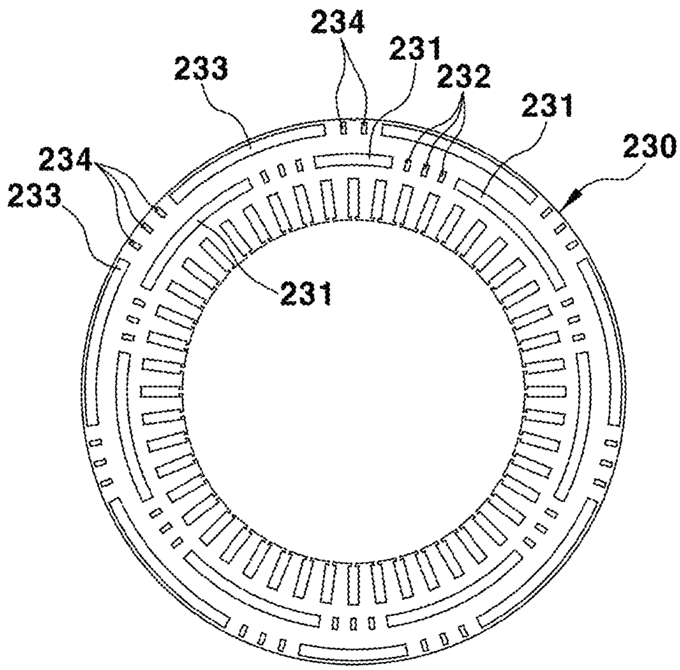
234
231
232
233
231
230
234
233
231

FIG. 6B

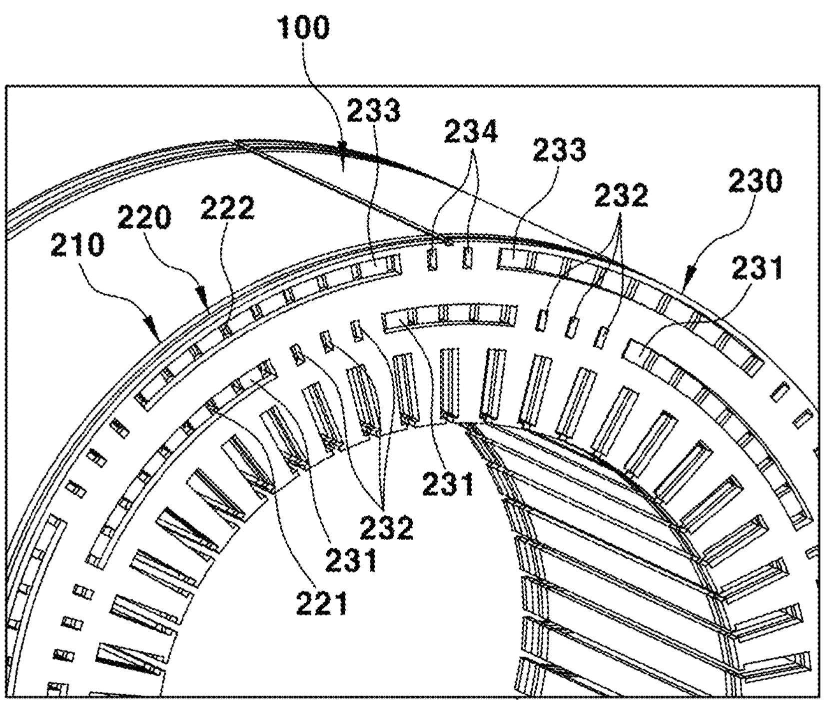
100
233
234
233
232
230
220
222
210
231
231
232
231
221

FIG. 7A

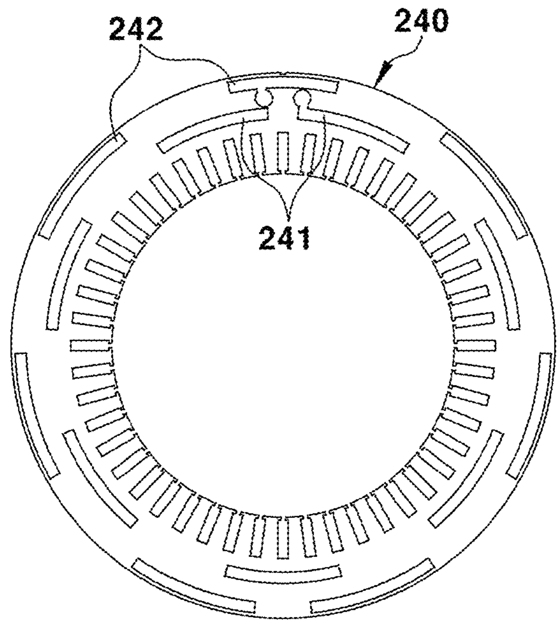
242
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FIG. 7B

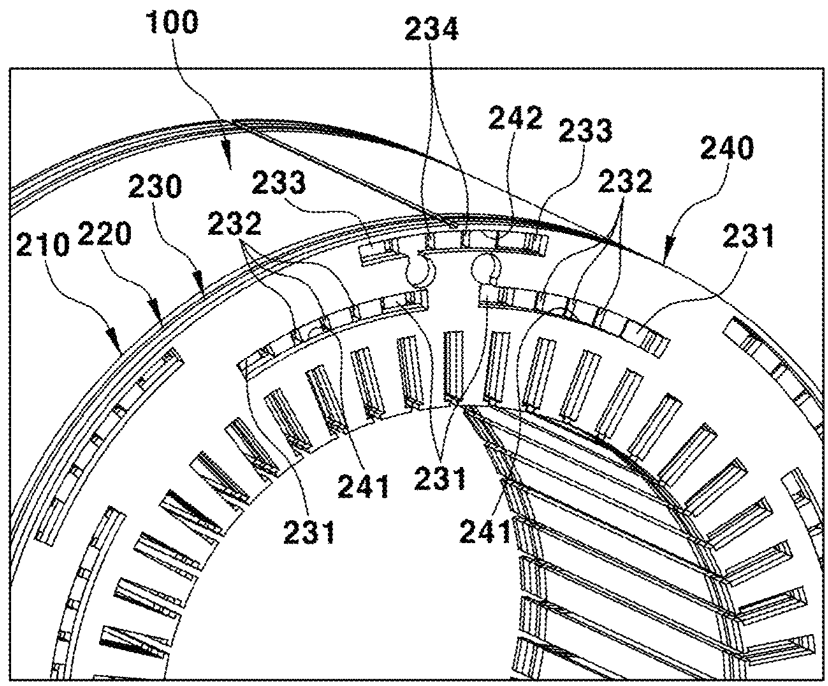

100
234
242
233
240
233
232
230
232
231
210
220
232
241
231
231
241

FIG. 8A

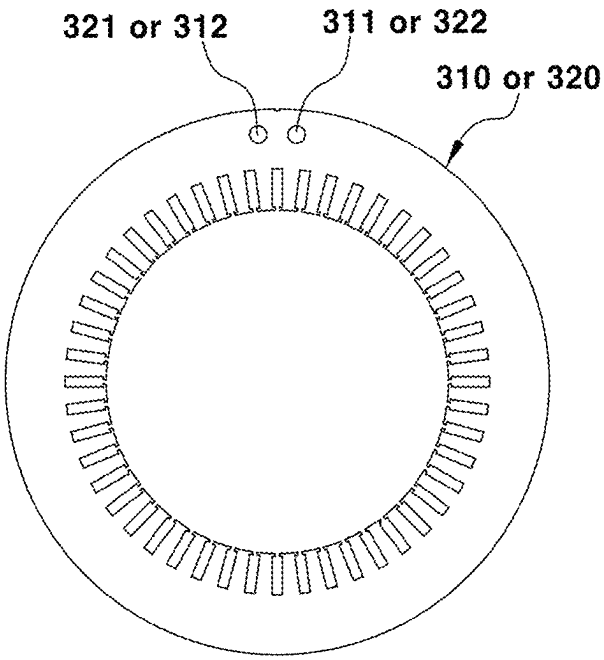

321 or 312
311 or 322
310 or 320

FIG. 8B

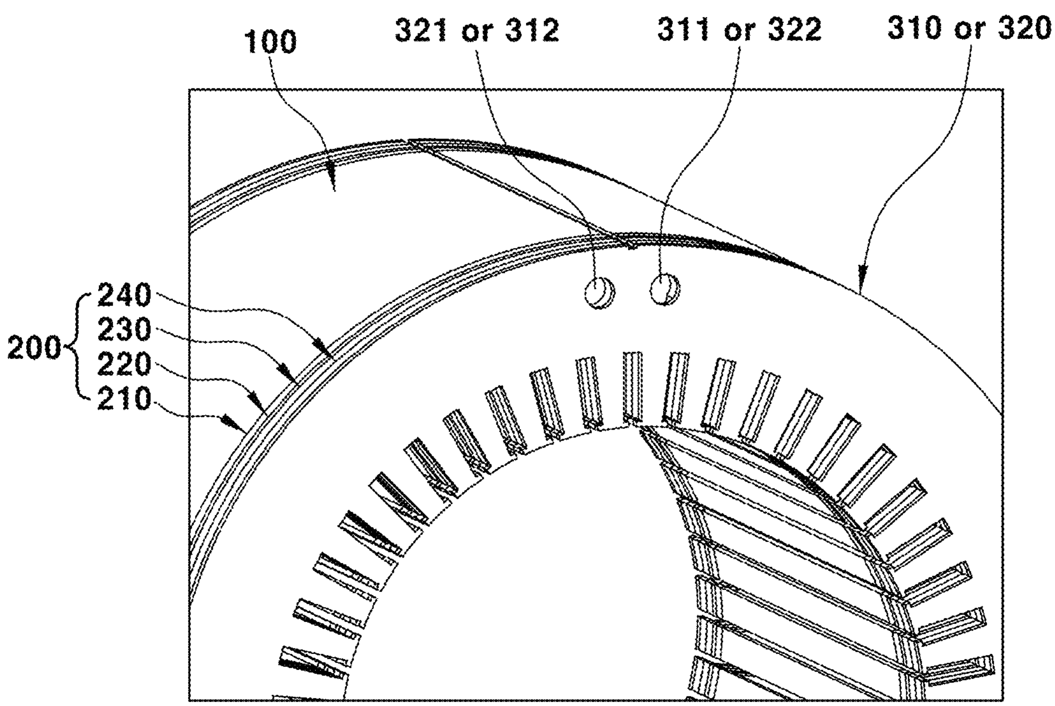

100
321 or 312
311 or 322
310 or 320
200
240
230
220
210

FIG. 9

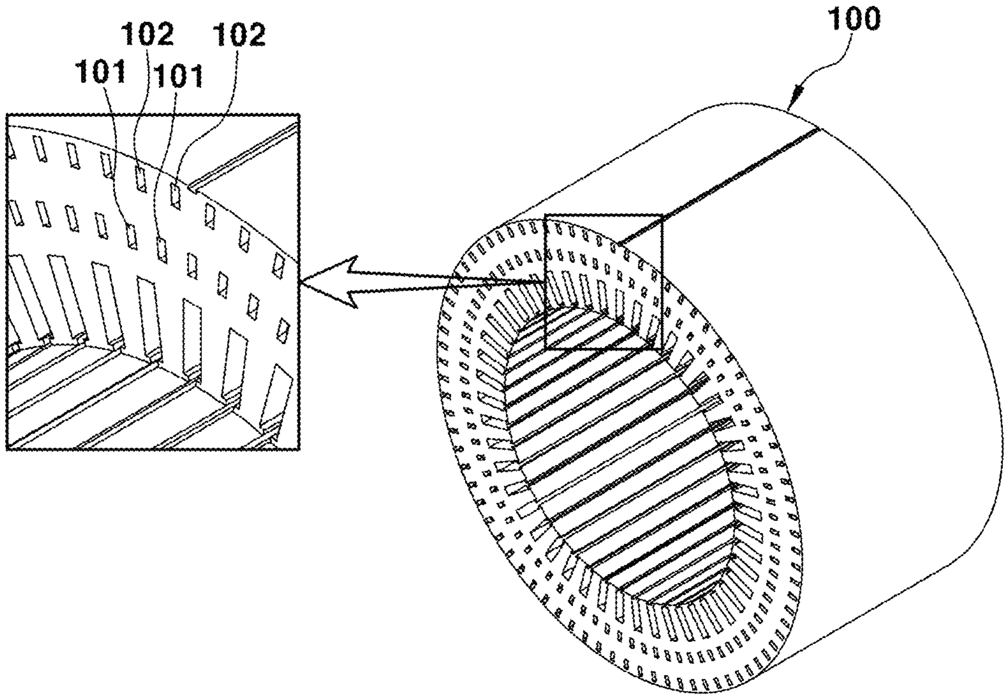


FIG. 10

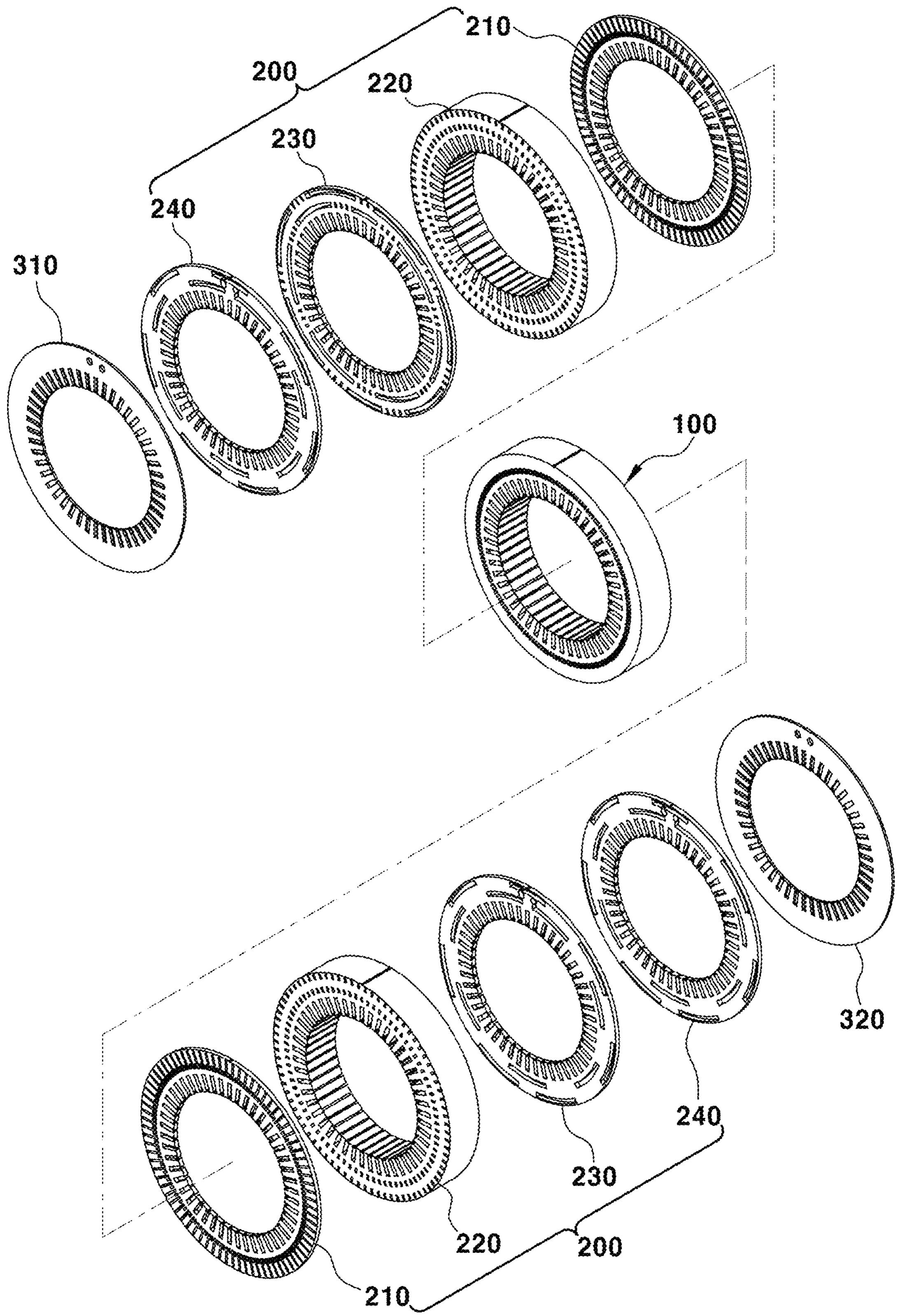
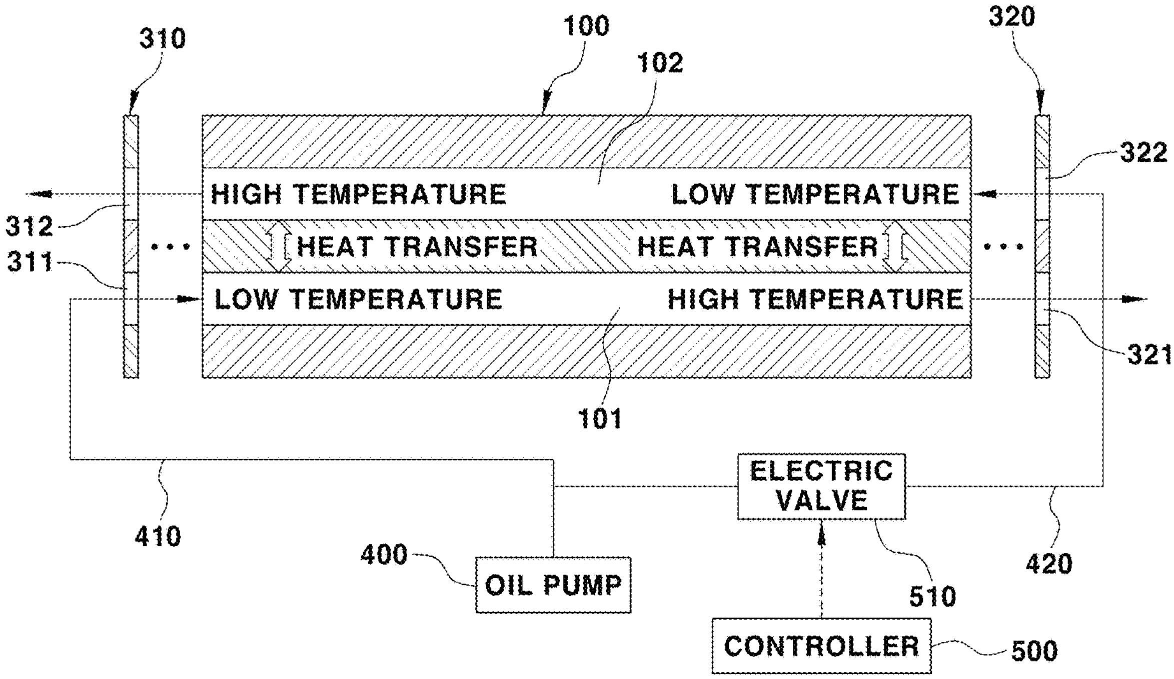


FIG. 11


310
100
102
320
322
HIGH TEMPERATURE
LOW TEMPERATURE
312
311
HEAT TRANSFER
HEAT TRANSFER
LOW TEMPERATURE
HIGH TEMPERATURE
321
101
ELECTRIC VALVE
410
420
400
OIL PUMP
510
CONTROLLER
500

DEVICE FOR COOLING A MOTOR

CROSS-REFERENCE TO RELATED APPLICATION

This application claims under 35 U.S.C. § 119(a) the benefit of priority to Korean Patent Application No. 10-2022-0044292 filed on Apr. 11, 2022, the entire contents of which are incorporated herein by reference.

BACKGROUND

(a) Technical Field

The present disclosure relates to a device for cooling a motor, and more particularly, to a device for cooling a motor, in which motor cooling performance can be maximized with multiple cooling channels formed in a stator core of the motor, allowing a coolant to flow in both directions therethrough.

(b) Background Art

Generally, environment-friendly vehicles such as electric vehicles, hybrid electric vehicles, and fuel cell vehicles are equipped with drive motors. Such driver motors may be synchronous motors, induction motors, or the like as driving sources.

Typically, the motor includes a stator unit provided with stator coils wound around a stator core including a number of layered steel plates. The motor also includes a rotor unit provided with an output shaft fastened to a rotor core including a number of layered steel plates.

The motor containing the stator unit and the rotor unit generates excessive heat due to induced currents and the like that causes motor performance degradation. Thus, it is desirable that proper cooling be performed.

FIG. **1** is a schematic view showing an example of a motor cooling structure of the related art.

As shown in FIG. **1**, a single cooling channel **22** configured with one coolant inlet **21** and one coolant outlet **23** is formed in a stator core **10** of the motor.

More specifically, the single cooling channel **22**, which is configured with the coolant inlet **21** and the coolant outlet **23** at both end portions thereof, may be formed in an outer circumferential portion of the stator core **10** in a zigzag array.

Accordingly, cooling of the motor containing the stator core can be performed in such a manner that the coolant flows first through the coolant inlet **21**, taking away heat from the stator core **10** for cooling while circulating back and forth along the single cooling channel **22**. Then the coolant flows out through the coolant outlet **23**.

However, the motor cooling structure of the related art has the following problems.

The single cooling channel **22** of the related art is inevitably very long as the zigzag array is adopted therewith for circulating the coolant back and forth. Consequently, the temperature of the coolant flowing into the single cooling channel **22** through the coolant inlet **21** increases gradually as the coolant moves away from the coolant inlet **21**, resulting in a severe temperature gradient of the coolant within a single cooling channel **22**. Thus, the motor cooling performance is degraded because the cooling of the motor is not evenly performed due to the temperature gradient of the coolant.

For further explanation, because the single cooling channel **22** of the related art is very long, when the coolant performs the motor cooling by taking away heat from the stator core **10** by circulating back and forth along the single cooling channel **22**, the temperature of the coolant increases gradually as the coolant moves away from the coolant inlet **21**. Thus, the cooling performance of the stator core near the coolant outlet **23** is substantially degraded compared to the cooling performance of the stator core near the coolant inlet **21**. Accordingly, the motor cooling performance is reduced resulting from uneven cooling of the motor due to the temperature gradient of the coolant.

SUMMARY

The present disclosure has been made in efforts to solve the above-mentioned problems. An object of the present disclosure is to provide a device for cooling a motor, wherein the device is capable of maximizing motor cooling performance in such a manner that two or more multiple cooling channels are formed in a stator core with separate coolant inlets and outlets. Coolant moves in a straight line toward different directions along each cooling channel to cool the stator core. Thus, the temperature gradient of the coolant is minimized to cool the entire area of the motor more evenly.

To achieve the objects, according to the present disclosure, a device for cooling a motor is provided. The device includes a stator core provided with a plurality of first cooling channels and a plurality of second cooling channels formed therethrough along a leftward-rightward direction while being arranged alternately along a circumferential direction of the stator core. The device also includes first and second final core plates layered on opposed or both lateral surfaces of the stator core, respectively, with a configuration in which one coolant inlet and one coolant outlet are formed thereon. The device further includes a coolant delivery core layered between one lateral surface of the stator core and the first final core plate and between the other lateral surface of the stator core and the second final core plate. The coolant delivery core has a configuration in which all of the plurality of first cooling channels communicates with the coolant inlet of the first final core plate and the coolant outlet of the second final core plate, and in which all of the plurality of second cooling channels communicates with the coolant inlet of the second final core plate and the coolant outlet of the first final core plate.

In one example, the plurality of first cooling channels and the plurality of second cooling channels may be alternately formed with an arrangement in an identical concentric circle along a circumferential direction of the stator core.

Alternatively, the plurality of first cooling channels and the plurality of second cooling channels may be alternately formed in non-identical concentric circles along a circumferential direction of the stator core. The plurality of second cooling channels may be formed closer to an outer diameter surface of the stator core than the plurality of first cooling channels.

Particularly, the coolant delivery core may include a first core plate provided with a plurality of 1-1 communication holes corresponding to the plurality of first cooling channels, respectively, and with a plurality of 1-2 communication holes corresponding to the plurality of second cooling channels, respectively. The plurality of 1-1 communication holes and 1-2 communication holes may be formed through the first core plate along a leftward-rightward direction and arranged alternately along a circumferential direction of the first core plate. The coolant delivery core may further include a second core plate provided with a plurality of 2-1 communication holes corresponding to the plurality of 1-1 communication holes, respectively, and with a plurality of 2-2 communication holes corresponding to the plurality of 1-2 communication holes, respectively. The plurality of 2-1 communication holes and 2-2 communication holes may be formed through the second core plate along a leftward-rightward direction and arranged alternately along a circumferential direction of the second core plate. The coolant delivery core may further include a third core plate provided with: a plurality of 3-1 slots communicating with some of the plurality of 2-1 communication holes, formed to be spaced at a predetermined distance along a circumferential direction thereof; a plurality of 3-1 communication holes corresponding to some of the plurality of 2-1 communication holes, respectively, formed between the 3-1 slots; a plurality of 3-2 slots communicating with some of the plurality of 2-2 communication holes, formed to be spaced at a predetermined distance along a circumferential direction thereof; and a plurality of 3-2 communication holes corresponding to some of the 2-2 communication holes, respectively, formed between the 3-2 slots. The coolant delivery core may also include a fourth core plate provided with a plurality of 4-1 slots communicating with neighboring 3-1 slots of the plurality of 3-1 slots and also communicating with one of the plurality of 3-1 communication holes between the neighboring 3-1 slots, formed to be spaced at a predetermined distance along a circumferential direction thereof. The fourth core plate may further include a plurality of 4-2 slots communicating with neighboring 3-2 slots of the plurality of 3-2 slots and also communicating with one of the plurality of 3-2 communication holes between the neighboring 3-2 slots, formed to be spaced at a predetermined distance along a circumferential direction thereof. The first final core plate or the second final core plate is layered on an outer surface of the fourth core plate for being combined therewith.

In one example, the plurality of 1-2 communication holes may be formed longer in a radial direction than the plurality of 1-1 communication holes.

In another example, the plurality of 2-2 communication holes may be formed closer to an outer diameter surface of the second core plate than the plurality of 2-1 communication hole.

In another example, the plurality of 3-2 slots and the plurality of 3-2 communication holes may be formed closer to an outer diameter surface of the third core plate than the plurality of 3-1 slots and the plurality of 3-1 communication holes.

In another example, the plurality of 4-2 slots may be formed closer to an outer diameter surface of the fourth core plate than the plurality of 4-1 slots.

In addition, one of the plurality of 4-1 slots may communicate with a coolant inlet of the first final core plate or a coolant outlet of the second final core plate. Additionally, one of the plurality of 4-2 slots may communicate with a coolant inlet of the second final core plate or a coolant outlet of the first final core plate.

The second core plate may be adopted with an increased thickness, or the stator core may be adopted with a reduced thickness in such a manner as to adjust the moving length of the coolant by increasing the leftward-rightward lengths of the plurality of 2-1 communication holes and the plurality of 2-2 communication holes or decreasing the leftward-rightward lengths of the first cooling channel and the second cooling channel.

In addition, the coolant inlet of the first final core plate and the coolant inlet of the second final core plate may be connected with a discharge part of a cooling fluid pump for pumping the coolant supply.

In addition, the coolant inlet of the first final core plate and the coolant inlet of the second final core plate may be connected with a first coolant supply line and a second coolant supply line, respectively, which are branched and extended from the discharge part of the cooling fluid pump.

In addition, the first coolant supply line or the second coolant supply line may be equipped with an electric valve being opened or closed by a control signal of a controller.

In one example, the controller may be configured to control the electric valve to be open when an operating point of the motor is at a full load area, or a temperature of the motor is a reference temperature or higher.

Through the above configuration, the present disclosure provides the following effects.

First, by forming two or more multiple cooling channels with separate coolant inlets and outlets in the stator core, the length of the cooling channel for the cooling of the motor containing the stator core can be reduced. Thus, the temperature gradient of the coolants flowing through the cooling channels can be minimized.

Second, by forming two or more multiple cooling channels with separate coolant inlets and outlets in the stator core and allowing the coolant to move in a straight line in different directions along each cooling channel to cool the stator core, the cooling of the motor containing the stator core can be performed quickly and efficiently. Furthermore, motor cooling performance can be maximized in such a manner that the entire area of the motor is evenly cooled.

It should be understood that the terms "automotive" or "vehicular" or other similar terms as used herein are inclusive of motor vehicles in general. Such motor vehicles may encompass passenger automobiles including sports utility vehicles (SUVs), buses, trucks, various commercial vehicles, watercraft including a variety of boats and ships, aircraft, and the like. Such motor vehicles may also include hybrid vehicles, electric vehicles, plug-in hybrid electric vehicles, hydrogen-powered vehicles, and other alternative fuel vehicles (e.g., fuels derived from resources other than petroleum). As referred to herein, a hybrid vehicle is an automobile that has two or more sources of power, such as for example vehicles that are both gasoline-powered and electric-powered.

The above and other features of the disclosure are discussed below.

BRIEF DESCRIPTION OF THE DRAWINGS

The above and other features of the present disclosure are described in detail with reference to certain examples thereof illustrated in the accompanying drawings, which are given herein below by way of illustration only, and thus are not limitative of the present disclosure, and wherein:

FIG. **1** is a schematic view showing an example of a motor cooling structure of the related art;

FIG. **2** is an exploded perspective view showing a device for cooling a motor according to the present disclosure;

FIG. **3** is an enlarged perspective view showing an embodiment of a stator core in the configuration of the device for cooling a motor according to the present disclosure;

FIG. **4**A is a front view showing a first core plate of the device for cooling a motor according to the present disclosure;

FIG. **4**B is a partial enlarged perspective view showing the first core plate of the device for cooling a motor and layered on the stator core according to the present disclosure;

FIG. **5**A is a front view showing a second core plate of the device for cooling a motor according to the present disclosure;

FIG. **5**B is a partial enlarged perspective view showing the second core plate of the device for cooling a motor and layered on the first core plate according to the present disclosure;

FIG. **6**A is a front view showing a third core plate of the device for cooling a motor according to the present disclosure;

FIG. **6**B is a partial enlarged perspective view showing the third core plate of the device for cooling a motor and layered on the second core plate according to the present disclosure;

FIG. **7**A is a front view showing a fourth core plate of the device for cooling a motor according to the present disclosure;

FIG. **7**B is a partial enlarged perspective view showing the fourth core plate of the device for cooling a motor and layered on the third core plate according to the present disclosure;

FIG. **8**A is a front view showing first and second final core plates of the device for cooling a motor according to the present disclosure;

FIG. **8**B is a partial enlarged perspective view showing the first and second final core plates of the device for cooling a motor layered on the fourth core plate according to the present disclosure;

FIG. **9** is an enlarged perspective view showing another embodiment of a stator core in the configuration of the device for cooling a motor according to the present disclosure;

FIG. **10** is a perspective view showing an example of thickness adjustment for the stator core and the second core plate in the configuration of the device for cooling a motor according to the present disclosure; and

FIG. **11** is a schematic view showing an example of a cooling fluid pump and an electric valve connected to the device for cooling a motor by a coolant supply line according to the present disclosure.

It should be understood that the appended drawings are not necessarily drawn to scale, presenting a somewhat simplified representation of various advantageous features illustrative of the basic principles of the disclosure. The specific design features of the present disclosure as disclosed herein, including, for example, specific dimensions, orientations, locations, and shapes will be determined by the particular intended application and use environment.

In the figures, reference numbers refer to the same or equivalent sections of the present disclosure throughout the several figures of the drawing.

DETAILED DESCRIPTION

It should be understood that the terms "comprise," "have," and "include," and variations thereof, when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of at least one other features, integers, steps, operations, elements, components, and/or groups thereof. When a component, device, element, or the like of the present disclosure is described as having a purpose or performing an operation, function, or the like, the component, device, or element should be considered herein as being "configured to" meet that purpose or to perform that operation or function.

It should be understood that, although the terms first, second, and the like may be used herein to describe various elements, these elements should not be limited by these terms. It should also be understood that singular forms are intended to include the plural forms as well, unless the context clearly dictates otherwise.

Hereinafter, advantageous embodiments of the present disclosure are described in detail with reference to the accompanying drawings.

FIG. **2** is an exploded perspective view showing a device for cooling a motor according to the present disclosure. FIG. **3** is an enlarged perspective view showing an embodiment of a stator core in the configuration of the device for cooling a motor according to the present disclosure.

As shown in FIG. **2**, coolant delivery cores **200** are layered on opposed, i.e., both sides of a stator core **100** of a motor. The coolant delivery cores **200** may have identical configurations to each other. A first final core plate **310** and a second final core plate **320** are layered on outer sides of the coolant delivery cores **200**, respectively, for being combined therewith.

As shown in FIG. **3**, the stator core **100** is provided with a configuration in which a plurality of first cooling channels **101** and a plurality of second cooling channels **102** are formed therethrough along a leftward-rightward direction and arranged alternately along a circumferential direction of the stator core **100**.

As shown in FIG. **3**, when a thickness of the stator core **100**, i.e., a cooling target, in a radial direction is less than a predetermined thickness, the plurality of first cooling channels **101** and the plurality of second cooling channels **102** may be alternately formed in one identical concentric circle along the circumferential direction of the stator core **100**.

Alternatively, as shown in FIG. **9**, when a thickness of the stator core **100**, i.e., the cooling target, in a radial direction is not less (i.e., equal to or greater) than the predetermined thickness, for smooth, even cooling of the stator core **100**, the plurality of first cooling channels **101** and the plurality of second cooling channels **102** may be alternately formed in non-identical concentric circles to each other along the circumferential direction of the stator core **100**. Thus, the plurality of second cooling channels **102** may be formed closer to an outer diameter surface of the stator core **100** than the plurality of first cooling channels **101**.

In this way, the coolant will flow in a straight line toward one direction along the first cooling channel **101** of the stator core **100**, or in a straight line toward the other direction along the second cooling channel **102** of the stator core **100**.

The first final core plate **310** and the second final core plate **320** are provided having identical or similar shapes to each other, and one coolant inlet and one coolant outlet are formed on each of the first and second final core plates **310**, **320**. For example, as shown in FIG. **8**A, one first coolant inlet **311** and one second coolant outlet **312** are formed at a predetermined position through the first final core plate **310**, and one first coolant outlet **321** and one second coolant inlet **322** are formed at a predetermined position through the second final core plate **320** therethrough.

The coolant delivery core **200** is layered between one lateral surface of the stator core **100** and the first final core plate **310** in such a manner as to deliver the coolant. The coolant delivery core **200** is also layered between the other lateral surface of the stator core **100** and the second final core plate **320** in such a manner as to deliver the coolant.

For this purpose, the coolant delivery core **200** is provided with a configuration in which all of the plurality of first cooling channels **101** communicate with the first coolant inlet **311** of the first final core plate **310** and the first coolant outlet **321** of the second final core plate **320**. The coolant delivery core **200** is also provided with a configuration in which all of the plurality of second cooling channels **102** communicate with the second coolant inlet **322** of the second final core plate **320** and the second coolant outlet **312** of the first final core plate **310**.

For further explanation, the coolant delivery core **200** is provided with a configuration in which the inflow coolant flows into one first coolant inlet **311** of the first final core plate **310** and is guided to flow through each of the plurality of first cooling channels **101** formed in the stator core **100**. Furthermore, all of the outflow coolant flows from each first cooling channel **101** and is guided to flow through one first coolant outlet **321** of the second final core plate **320**.

In addition, the coolant delivery core **200** is provided with a configuration in which the inflow coolant flows into one second coolant inlet **322** of the second final core plate **320** and is guided to flow through each of the plurality of second cooling channels **102** formed in the stator core **100**. Furthermore, all of the outflow coolant flows from each second cooling channel **102** and is guided to flow through one second coolant outlet **312** of the first final core plate **310**.

To this end, the coolant delivery core **200** may include first to fourth core plates **210**, **220**, **230**, and **240** that are layered between one lateral surface of the stator core **100** and the first final core plate **310**, and between the other lateral surface of the stator core **100** and the second final core plate **320**.

As shown in FIGS. **4**A and **4**B, the first core plate **210** may be provided with a configuration in which a plurality of 1-1 communication holes **211** correspond to the plurality of first cooling channels **101**, respectively, and in which a plurality of 1-2 communication holes **212** correspond to the plurality of second cooling channels **102**, respectively. The plurality of 1-1 communication holes **211** and 1-2 communication holes **212** are formed through the first core plate **210** along a leftward-rightward direction and arranged alternately along a circumferential direction of the first core plate **210**.

The plurality of 1-2 communication holes **212** is formed longer in a radial direction of the first core plate **210** than the plurality of 1-1 communication holes **211**.

As shown in FIGS. **5**A and **5**B, the second core plate **220** may be provided with a configuration in which a plurality of 2-1 communication holes **221** correspond to the plurality of 1-1 communication holes **211**, respectively, and in which a plurality of 2-2 communication holes **222** correspond to the plurality of 1-2 communication holes **212**, respectively. The plurality of 2-1 communication holes **221** and 2-2 communication holes **222** are formed through the second core plate **220** along a leftward-rightward direction and arranged alternately along a circumferential direction of the second core plate **220**.

The plurality of 2-2 communication holes **222** is formed closer to an outer diameter surface of the second core plate **220** than the plurality of 2-1 communication holes **221**.

As shown in FIGS. **6**A and **6**B, the third core plate **230** may be provided with a configuration in which a plurality of 3-1 slots **231** communicate with some of the plurality of 2-1 communication holes **221** and are formed to be spaced at a predetermined distance along a circumferential direction thereof. The third core plate **230** may be also provided with a plurality of 3-1 communication holes **232** that correspond to some of the plurality of 2-1 communication holes **221**, respectively and are formed between the 3-1 slots **231**. Furthermore, the third core plate **230** may be provided with a plurality of 3-2 slots **233** that communicate with some of the plurality of 2-2 communication holes **222** and are formed to be spaced at a predetermined distance along a circumferential direction thereof. Additionally, the third core plate **230** may be provided with a plurality of 3-2 communication holes **234** that correspond to some of the plurality of 2-2 communication holes **222**, respectively, and are formed between the plurality of 3-2 slots **233**.

The plurality of 3-1 slots **231** and 3-1 communication holes **232** are arranged with each other on one identical concentric circle, and the plurality of 3-2 slots **233** and 3-2 communication holes **234** are also arranged with each other on another identical concentric circle. Additionally, the plurality of 3-2 slots **233** and 3-2 communication holes **234** are formed closer to an outer diameter surface of the third core plate **230** than the plurality of 3-1 slots **231** and 3-1 communication holes **232**.

As shown in FIGS. **7**A and **7**B, the fourth core plate **240** may be provided with a configuration in which a plurality of 4-1 slots **241** is formed to be spaced at a predetermined distance along a circumferential direction thereof, and a plurality of 4-2 slots **242** is formed to be spaced at a predetermined distance along a circumferential direction thereof. The plurality of 4-1 slots **241** communicates with two neighboring 3-1 slots **231** of the plurality of 3-1 slots **231** and also communicates with the 3-1 communication hole **232** disposed between the two neighboring 3-1 slots **231**. Furthermore, the plurality of 4-2 slots **242** communicates with two neighboring 3-2 slots **233** of the plurality of 3-2 slots **233** and also communicates with the 3-2 communication hole **234** between the two neighboring 3-2 slots **233**.

The plurality of 4-1 slots **241** is arranged on one identical concentric circle, and the plurality of 4-2 slots **242** are also arranged on another identical concentric circle. Additionally, the plurality of 4-2 slots **242** is formed closer to an outer diameter surface of the fourth core plate **240** than the plurality of 4-1 slots **241**.

First to fourth core plates **210**, **220**, **230**, and **240** are layered in order on one lateral surface and on the other opposite lateral surface of the stator core **100**. The first final core plate **310** is layered on an outer surface of the fourth core plate **240** on one lateral side of the core **100** and the second final core plate **320** is layered on an outer surface on the other opposite lateral side of the fourth core plate **240** for being combined therewith.

In addition, as shown in FIGS. **7**B and **8**B, when the first final core plate **310** or the second final core plate **320** is layered on the outer surface of the fourth core plate **240** for being combined therewith, one of the plurality of 4-1 slots **241** fluidly communicates with the first coolant inlet **311** of the first final core plate **310** or the first coolant outlet **321** of the second final core plate **320**. Additionally, when the first final core plate **310** or the second final core plate **320** is layered on the outer surface of the fourth core plate **240** for being combined therewith, one of the plurality of 4-2 slots **242** fluidly communicates with the second coolant inlet **322** of the second final core plate **320** or the second coolant outlet **312** of the first final core plate **310**.

As shown in FIG. **11**, the first coolant inlet **311** of the first final core plate **310** and the second coolant inlet **322** of the second final core plate **320** are connected with a discharge part of a cooling fluid pump **400**. The cooling fluid pump **400** supplies the coolant by pumping the coolant through the first inlet **311** of the first final core plate **310** and the second coolant inlet **322** of the second final core plate **320**.

More specifically, the first coolant inlet **311** of the first final core plate **310** and the second coolant inlet **322** of the second final core plate **320** are connected with a first coolant supply line **410** and a second coolant supply line **420**, respectively. The first coolant supply line **410** and the second coolant supply line **420** branch and extend from the discharge part of the cooling fluid pump **400**.

In this way, the coolant (or cooling oil) may be supplied to the first coolant inlet **311** of the first final core plate **310** along the first coolant supply line **410** by the pumping drive of the cooling fluid pump **400**. The coolant (or cooling oil) may also be supplied to the second coolant inlet **322** of the second final core plate **320** along the second coolant supply line **420** by the pumping drive of the cooling fluid pump **400**.

In this case, an electric valve **510** that is opened or closed by a control signal of a controller **500** may be installed on the first coolant supply line **410** or the second coolant supply line **420**.

The electric valve **510** that is opened or closed by a control signal of the controller **500** may only be installed on the second coolant supply line **420**.

Accordingly, the electric valve **510** is normally closed, and the coolant (or cooling oil) is only supplied to the first coolant inlet **311** of the first final core plate **310** along the first coolant supply line **410** by the pumping drive of the cooling fluid pump **400**.

In contrast, when the controller **500** determines that an operating point of the motor is at a full load area, or a temperature of the motor is a reference temperature or higher, the electric valve **510** is controlled to open. Thus, the coolant (or cooling oil) may be supplied to the first coolant inlet **311** of the first final core plate **310** along the first coolant supply line **410** and also to the second coolant inlet **322** of the second final core plate **320** along the second coolant supply line **420** by the pumping drive of the cooling fluid pump **400**.

When a specification of the stator core varies depending on the motor type, the leftward-rightward lengths of the first cooling channel **101** and the second cooling channel **102** in the stator core **100** may vary accordingly. Thus, the motor cooling path for the coolant flow may vary and the thickness of the second core plate **220** may be increased or decreased depending on the leftward-rightward thickness of the stator core **100**.

For example, as shown in FIG. **10**, considering that the motor cooling path for the coolant flow may vary, the second core plate **220** may be adopted with an increased thickness, and the stator core **100** may be adopted with a reduced thickness.

In this way, the leftward-rightward lengths of the plurality of 2-1 communication holes **221** and 2-2 communication holes **222** in the second core plate **220** may be increased by adopting the second core plate **220** with an increased thickness. Also, the leftward-rightward lengths of the first cooling channel **101** and the second cooling channel **102** in the stator core **100** may be decreased by adopting the stator core **100** with a reduced thickness. Eventually, the moving length of the coolant, as the motor cooling path for the coolant flow, may be adjusted.

The coolant flow process performed in the device for cooling a motor of the present disclosure, including the above-described configurations, is reviewed as follows.

The coolant (or cooling oil) passes through the first coolant supply line **410**, the first final core plate **310**, the first cooling channel **101** of the coolant delivery core **200** that is layered between the first final core plate **310** and one lateral surface of the stator core **100**, and the second final core plate **320** sequentially by the pumping drive of the cooling fluid pump **400**. Thus, the cooling of the motor containing the stator core **100** may be performed.

To this end, the coolant (or cooling oil) is first supplied to the first coolant inlet **311** of the first final core plate **310** along the first coolant supply line **410** by the pumping drive of the cooling fluid pump **400**. Thus, the cooling of the motor containing the stator core **100** is performed while the coolant flows in a straight line toward one direction along the first cooling channel **101** formed in the stator core **100**.

Also, as shown in FIGS. **7**B and **8**B, the coolant supplied to the first coolant inlet **311** of the first final core plate **310** may flow into one 4-1 slot **241**, which communicates with the first coolant inlet **311** of the plurality of 4-1 slots **241** formed in the fourth core plate **240**.

Next, the coolant that flows into the one 4-1 slot **241** may flow through the corresponding 3-1 slot **231** and 3-1 communication hole **232** of the third core plate **230**.

More specifically, as shown in FIG. **7**B, in addition to the one 4-1 slot **241** communicating with the first coolant inlet **311**, the remaining 4-1 slots **241** communicate all at once with two neighboring 3-1 slots **231** of the plurality of 3-1 slots **231** formed in the third core plate **230** and also communicate with the 3-1 communication hole **232** between the two neighboring 3-1 slots **231**. Thus, the coolant that flows into the 4-1 slot **241** may easily flow into the 3-1 slot **231** and the 3-1 communication hole **232** of the third core plate **230**.

The coolant that flows into the 3-1 slot **231** and the 3-1 communication hole **232** of the third core plate **230** may successively flow into the 2-1 communication hole **221** of the second core plate **220**.

More specifically, as shown in FIG. **6**B, the 3-1 slot **231** of the third core plate **230** communicates all at once with some of the plurality of 2-1 communication holes **221** formed in the second core plate **220**, and the 3-1 communication holes **232** of the third core plate **230** communicate one-to-one with some of the plurality of 2-1 communication holes **221** formed in the second core plate **220**. Thus, the coolant that flows into the 3-1 slot **231** and the 3-1 communication hole **232** of the third core plate **230** may easily flow into each of the plurality of 2-1 communication holes **221** formed in the second core plate **220**.

Then, the coolant that flows into the plurality of 2-1 communication holes **221** formed in the second core plate **220** may flow into the plurality of 1-1 communication holes **211** formed in the first core plate **210**.

For further explanation, as shown in FIG. **5**B, the 2-1 communication holes **221** of the second core plate **220** communicate one-to-one with the 1-1 communication holes **211** of the first core plate **210**. Thus, the coolant that flows into the 2-1 communication holes **221** of the second core plate **220** may easily flow into the 1-1 communication holes **211** of the first core plate **210**.

Next, the coolant that flows into the 1-1 communication holes **211** of the first core plate **210** may flow into the first cooling channel **101** of the stator core **100**.

For further explanation, as shown in FIG. **4**B, the 1-1 communication holes **211** of the first core plate **210** communicate one-to-one with the plurality of first cooling channels **101** formed in the stator core **100**. Thus, the coolant that flows into the 1-1 communication holes **211** of the first core plate **210** may flow into the first cooling channel **101** of the stator core **100**.

Therefore, the coolant that flows into the first cooling channel **101** of the stator core **100** may perform the cooling of the motor containing the stator core **100** while flowing in a straight line toward one direction.

In this way, the coolant supplied to the first coolant inlet **311** of the first final core plate **310** may perform the cooling of the motor containing the stator core **100** while flowing in a straight line toward one direction upon coming in the first cooling channel **101** of the stator core **100** after passing through the coolant delivery core **200**. The coolant delivery core **200** is arranged between the first final core plate **310** and one lateral surface of the stator core **100**, i.e., after passing through the fourth core plate **240**, the third core plate **230**, the second core plate **220**, and the first core plate **210** in order.

In this case, the coolant that flows out of the first cooling channel **101** of the stator core **100**, after passing through the identical coolant delivery core **200** arranged between the other lateral surface of the stator core **100** and the second final core plate **320**, may be discharged through the first coolant outlet **321** of the second final core plate **320**.

For further explanation, the coolant that flows out of the first cooling channel **101** of the stator core **100** may be discharged through the first coolant outlet **321** of the second final core plate **320** after passing through the first core plate **210**, the second core plate **220**, the third core plate **230**, and the fourth core plate **240** in order. The first to fourth core plates **210**, **220**, **230**, and **240** are arranged between the other lateral surface of the stator core **100** and the second final core plate **320**. In other words, the first to fourth core plates **210**, **220**, **230**, and **240** are in reverse order when the coolant flows through the fourth core plate **240**, the third core plate **230**, the second core plate **220**, and the first core plate **210** in the order as described herein.

When the controller **500** determines that an operating point of the motor is at a full load area, or a temperature of the motor is at a reference temperature or higher, the electric valve **510** is controlled to open, and thus the coolant (or cooling oil) may also be supplied to the second coolant inlet **322** of the second final core plate **320** along the second coolant supply line **420** by the pumping drive of the cooling fluid pump **400**.

In other words, the coolant (or cooling oil) may be supplied to the second coolant inlet **322** of the second final core plate **320** along the second coolant supply line **420** by the pumping drive of the cooling fluid pump **400**. Thus, the cooling of the motor containing the stator core **100** is further performed while the coolant flows in a straight line toward the other direction along the second cooling channel **102** formed in the stator core **100**.

In this way, the coolant (or cooling oil) passes through the second coolant supply line **420**, the second final core plate **320**, the second cooling channel **102** of the coolant delivery core **200** that is layered between the second final core plate **320** and the other lateral surface of the stator core **100**, and the first final core plate **310** sequentially by the pumping drive of the cooling fluid pump **400**. Thus, the cooling of the motor containing the stator core **100** may be maximized.

To this end, the coolant (or cooling oil) is first supplied to the second coolant inlet **322** of the second final core plate **320** along the second coolant supply line **420** by the pumping drive of the cooling fluid pump **400** so that the cooling of the motor containing the stator core **100** is further performed while the coolant flows in a straight line toward the other direction along the second cooling channel **102** formed in the stator core **100**.

Then, as shown in FIGS. **7**B and **8**B, the coolant supplied to the second coolant inlet **322** of the second final core plate **320** may flow into one 4-2 slot **242**, which communicates with the second coolant inlet **322** of the plurality of 4-2 slots **242** formed in the fourth core plate **240**.

Next, the coolant that flows into the one 4-2 slot **242** may flow through the corresponding 3-2 slot **233** and the 3-2 communication hole **234** of the third core plate **230**.

More specifically, as shown in FIG. **7**B, in addition to the one 4-2 slot **242** communicating with the second coolant inlet **322**, the remaining 4-2 slots **242** communicate all at once with two neighboring 3-2 slots **233** out of the plurality of 3-2 slots **233** formed in the third core plate **230** and also communicate with the 3-2 communication hole **234** between the two neighboring 3-2 slots **233**. Thus, the coolant that flows into the 4-2 slot **242** may easily flow into the 3-2 slot **233** and a 3-3 communication hole **234** of the third core plate **230**.

The coolant that flows into the 3-2 slots **233** and the 3-3 communication holes **234** of the third core plate **230** may successively flow into the 2-2 communication holes **222** of the second core plate **220**.

More specifically, as shown in FIG. **6**B, each 3-2 slot **233** of the third core plate **230** communicates all at once with some of the plurality of 2-2 communication holes **222** formed in the second core plate **220**. The 3-2 communication holes **234** of the third core plate **230** communicate one-to-one with some of the plurality of 2-2 communication holes **222** formed in the second core plate **220**. Thus, the coolant that flows into the 3-2 slot **233** and the 3-2 communication hole **234** of the third core plate **230** may easily flow into each of the plurality of 2-2 communication holes **222** formed in the second core plate **220**. Then, the coolant that flows into the plurality of 2-2 communication holes **222** formed in the second core plate **220** may flow into the plurality of 1-2 communication holes **212** formed in the first core plate **210**.

For further explanation, as shown in FIG. **5**B, the 2-2 communication holes **222** of the second core plate **220** communicate one-to-one with the 1-2 communication holes **212** of the first core plate **210**. Thus, the coolant that flows into the 2-2 communication holes **222** of the second core plate **220** may easily flow into the 1-2 communication holes **212** of the first core plate **210**.

Next, the coolant that flows into the 1-2 communication holes **212** of the first core plate **210** may flow into the second cooling channel **102** of the stator core **100**.

For further explanation, as shown in FIG. **4**B, the 1-2 communication holes **212** of the first core plate **210** communicate one-to-one with the plurality of second cooling channels **102** formed in the stator core **100**. Thus, the coolant that flows into the 1-2 communication holes **212** of the first core plate **210** may flow into the second cooling channel **102** of the stator core **100**.

Therefore, the coolant that flows into the second cooling channel **102** of the stator core **100** may perform the cooling of the motor containing the stator core **100** while flowing in a straight line toward the other direction.

In this way, the coolant supplied to the second coolant inlet **322** of the second final core plate **320** may perform the cooling of the motor containing the stator core **100** while flowing in a straight line toward the other direction upon coming into the second cooling channel **102** of the stator core **100** after passing through the coolant delivery core **200**. The coolant delivery core **200** is arranged between the second final core plate **320** and the other lateral surface of the stator core **100**, i.e., after passing through the fourth core plate **240**, the third core plate **230**, the second core plate **220**, and the first core plate **210** in order.

In this case, the coolant that flows out of the second cooling channel **102** of the stator core **100**, after passing through the identical coolant delivery core **200** arranged between one lateral surface of the stator core **100** and the first final core plate **310**, may be discharged through the second coolant outlet **312** of the first final core plate **310**.

For further explanation, the coolant that flows out of the second cooling channel **102** of the stator core **100** may be discharged through the second coolant outlet **312** of the first final core plate **310** after passing through the first core plate **210**, the second core plate **220**, the third core plate **230**, and the fourth core plate **240** in order. The first to fourth core plates **210**, **220**, **230**, and **240** are arranged between one lateral surface of the stator core **100** and the first final core plate **310**, i.e., in reverse order when the coolant flows through the fourth core plate **240**, the third core plate **230**, the second core plate **220**, and the first core plate **210** in order as described herein.

In this way, when the controller **500** determines that an operating point of the motor is at a full load area, or a temperature of the motor is a reference temperature or higher, the electric valve **510** is controlled to open. This enables the coolant (or cooling oil) to cool the stator core by the pumping drive of the cooling fluid pump **400** while the coolant flows in a straight line toward one side along the first cooling channel **101** of the stator core **100** and also flows in a straight line toward the other side along the second cooling channel **102** at the same time. Thus, the motor cooling can be maximized.

Furthermore, referring to FIG. **11**, the inflow coolant into the first cooling channel **101** of the stator core **100** through the first coolant inlet **311** of the first final core plate **310** is in a low-temperature state, and the outflow coolant from the second cooling channel **102** of the stator core **100** through the first coolant outlet **312** of the first final core plate **320** is in a high-temperature state. Thus, the temperature gradient of the coolant flowing through each of the cooling channels **101** and **102** can be minimized by the heat transfer between the low-temperature coolant flowing through the first cooling channel **101** and the high-temperature coolant flowing through the second cooling channel **102**.

Likewise, as shown in FIG. **11**, the inflow coolant into the second cooling channel **102** of the stator core **100** through the second coolant inlet **322** of the second final core plate **320** is in a low-temperature state, and the outflow coolant from the first cooling channel **101** of the stator core **100** through the second coolant outlet **321** of the second final core plate **320** is in a high-temperature state. Thus, the temperature gradient of the coolants flowing through each of the cooling channels **101** and **102** can be minimized by the heat transfer between the low-temperature coolant flowing through the second cooling channel **102** and the high-temperature coolant flowing through the first cooling channel **101**.

As described above, two or more multiple cooling channels **101** and **102** with separate coolant inlets and outlets are formed in the stator core **100** to reduce the length of the cooling channel for the cooling of the motor containing the stator core **100**. Thus, the temperature gradient of the coolants flowing through the cooling channels can be minimized.

In addition, two or more multiple cooling channels **101** and **102** are formed in the stator core **100** with separate coolant inlets and outlets, allowing the coolant to move along each cooling channel in a straight line in different directions to cool the stator core **100**. Thus, the cooling of the motor containing the stator core can be performed quickly and efficiently, and motor cooling performance can be maximized in such a manner that the entire area of the motor is evenly cooled and the like.

While various embodiments of the present disclosure have been described herein in detail, the scope of the present disclosure is not limited to each of the described embodiments. Furthermore, various modifications and improvements by those having ordinary skill in the art, using the basic concept of the present disclosure defined in the appended claims, are also included in the scope of the present disclosure.

What is claimed is:

1. A device for cooling a motor, the device comprising:

a stator core with a plurality of first cooling channels and a plurality of second cooling channels formed therethrough along an axial direction and arranged alternately along a circumferential direction of the stator core; and

first and second final core plates layered, respectively, on opposed lateral surfaces of the stator core, each having a configuration of one coolant inlet and one coolant outlet formed therein;

wherein the stator core includes a first coolant delivery core disposed on a left side thereof and capped with the first final core plate and a second coolant delivery core disposed on a right side thereof and capped with the second final core plate,

wherein the plurality of first cooling channels all communicates between a coolant inlet of the first final core plate and a coolant outlet of the second final core plate, and the plurality of second cooling channels all communicate between a coolant inlet of the second final core plate and a coolant outlet of the first final core plate,

wherein each of the first coolant delivery core and the second coolant delivery core comprises:

a first core plate provided with a plurality of 1-1 communication holes corresponding to the plurality of first cooling channels, respectively, and a plurality of 1-2 communication holes corresponding to the plurality of second cooling channels, respectively, formed therethrough along a leftward-rightward direction while being arranged alternately along a circumferential direction of the first core plate;

a second core plate provided with a plurality of 2-1 communication holes corresponding to the plurality of 1-1 communication holes, respectively, and with a plurality of 2-2 communication holes corresponding to the plurality of 1-2 communication holes, respectively, formed therethrough along a leftward-rightward direction while being arranged alternately along a circumferential direction of the second core plate;

a third core plate provided with a plurality of 3-1 slots communicating with some of the plurality of 2-1 communication holes, formed to be spaced at a predetermined distance along a circumferential direction thereof, 3-1 communication holes corresponding to some of the plurality of 2-1 communication holes, respectively, formed between the 3-1 slots, a plurality of 3-2 slots communicating with some of the plurality of 2-2 communication holes, formed to be spaced at a predetermined distance along a circumferential direction thereof, and 3-2 communication holes corresponding to some of the 2-2 communication holes, respectively, formed between the 3-2 slots; and a fourth core plate provided with a plurality of 4-1 slots communicating with neighboring 3-1 slots of the plurality of 3-1 slots and also communicating with a 3-1 communication hole between the neighboring 3-1 slots, formed to be spaced at a predetermined distance along a circumferential direction thereof, and a plurality of 4-2 slots communicating with neighboring 3-2 slots of the plurality of 3-2 slots and also communicating with a 3-2 communication hole between the neighboring 3-2 slots, formed to be spaced at a predetermined distance along a circumferential direction thereof,

wherein the first final core plate or the second final core plate is layered on an outer surface of the fourth core plate for being combined therewith.

2. The device of claim **1**, wherein the plurality of first cooling channels and the plurality of second cooling channels are arranged alternately in identical concentric circles along a circumferential direction of the stator core.

3. The device of claim **1**, wherein the plurality of first cooling channels and the plurality of second cooling channels are arranged alternately in non-identical concentric circles along a circumferential direction of the stator core.

4. The device of claim **3**, wherein the plurality of second cooling channels is formed closer to an outer diameter surface of the stator core than the plurality of first cooling channels.

5. The device of claim **1**, wherein each of the plurality of 1-2 communication holes has an opening with a diameter, in a radial direction, larger than that of each of the plurality of 1-1 communication holes.

6. The device of claim **1**, wherein the plurality of 2-2 communication holes is formed closer to an outer diameter surface of the second core plate than the plurality of 2-1 communication holes.

7. The device of claim **1**, wherein each 3-2 slot of the plurality of 3-2 slots and the 3-2 communication holes are formed closer to an outer diameter surface of the third core plate than the 3-1 slot and the 3-1 communication holes.

8. The device of claim **1**, wherein each 4-2 slot of the plurality of 4-2 slots is formed closer to an outer diameter surface of the fourth core plate than the plurality of 4-1 slots.

9. The device of claim **1**, wherein one of the plurality of 4-1 slots communicates with a coolant inlet of a first final core plate or a coolant outlet of a second final core plate, and one of the plurality of 4-2 slots communicates with a coolant inlet of the second final core plate or a coolant outlet of the first final core plate.

10. The device of claim **1**, wherein the second core plate has an increased thickness, or the stator core has a reduced thickness to adjust a flow length of the coolant by increasing leftward-rightward lengths of the plurality of 2-1 communication holes and 2-2 communication holes or decreasing leftward-rightward lengths of the first cooling channel and the second cooling channel.

11. The device of claim **1**, wherein the coolant inlet of the first final core plate and the coolant inlet of the second final core plate are connected with a discharge part of a cooling fluid pump configured to pump the coolant to supply the coolant to the stator core.

12. The device of claim **11**, wherein the coolant inlet of the first final core plate and the coolant inlet of the second final core plate are connected with a first coolant supply line and a second coolant supply line, respectively, which branch and extend from the discharge part of the cooling fluid pump.

13. The device of claim **12**, wherein the first coolant supply line or the second coolant supply line is equipped with an electric valve configured to be opened or closed by a control signal of a controller.

14. The device of claim **13**, wherein the controller is configured to control the electric valve to be open when an operating point of the motor is at a full load area, or a temperature of the motor is a reference temperature or higher.

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